

AI-Powered Intelligent Home Planning & Optimization System

AI-Based Vastu Detection & Compliance Evaluation Module

Group ID : 25-26J-201

BSc (Hons) degree in Information Technology
Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology
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Dissertation submitted in partial fulfillment of the requirements for the Bachelor of
Science (Hons) in Information Technology
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April 2026

DECLARATION OF THE CANDIDATE

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ABSTRACT

Vastu Shastra is a traditional architectural science that focuses on the spatial arrangement and directional alignment of structures to promote well-being, prosperity, and harmony. However, in modern construction practices, evaluating building layouts according to Vastu principles is often a complex and manual process that requires expert knowledge. This creates a need for an automated and intelligent system that can assist users in assessing Vastu compliance efficiently. This research proposes an AI-based Vastu analysis system that utilizes computer vision and rule-based evaluation techniques to analyze residential floor plans. The system accepts floor plan images as input and employs a YOLO (You Only Look Once) object detection model to identify different room types such as bedrooms, kitchens, and bathrooms. Once the rooms are detected, their spatial positions are mapped to corresponding directions within the layout. A rule-based engine is then applied to evaluate each detected element against predefined Vastu principles, such as proper placement of kitchens, avoidance of bathrooms in the northeast direction, and maintaining openness in specific zones. The system generates a structured analysis report indicating whether the layout complies with Vastu guidelines, along with explanations and recommendations. The proposed system aims to bridge the gap between traditional architectural knowledge and modern artificial intelligence technologies by providing a user-friendly and automated solution. Experimental observations indicate that the system can effectively detect room configurations and provide reliable Vastu evaluations, making it useful for homeowners, architects, and real estate planners.

Keywords: Vastu Shastra, computer vision, object detection, YOLO, floor plan analysis, rule-based system

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
ANN	Artificial Neural Network
CNN	Convolutional Neural Network
DNN	Deep Neural Network
YOLO	You Only Look Once
NMS	Non-Maximum Suppression
CLAHE	Contrast Limited Adaptive Histogram Equalization
IoU	Intersection over Union
mAP	Mean Average Precision
mAP@50	Mean Average Precision at IoU threshold 0.50
mAP@50–95	Mean Average Precision across IoU thresholds (0.50–0.95)
GPU	Graphics Processing Unit
API	Application Programming Interface
UI	User Interface
SaaS	Software as a Service
AEC	Architecture, Engineering, and Construction
BIM	Building Information Modeling
OpenCV	Open Source Computer Vision Library
T4	Tesla T4 GPU

1. INTRODUCTION

The rapid growth of urbanization and increasing population density have significantly impacted residential planning and architectural design. Modern housing development is constrained by limited land availability, irregular plot dimensions, and the need to accommodate both functional and cultural requirements. Traditional house planning processes rely heavily on manual expertise from architects and Vastu consultants, making them time-consuming, expensive, and often inconsistent.

Artificial Intelligence (AI) has emerged as a transformative technology capable of automating complex tasks in multiple domains, including image processing, spatial analysis, and decision support systems. In the context of residential planning, AI techniques such as Convolutional Neural Networks (CNNs) and object detection models have been widely used for floor plan analysis and room classification [3], [4]. Among these, the YOLO (You Only Look Once) algorithm has gained prominence due to its real-time detection capabilities and efficiency in identifying objects within images [6], [7].

Existing research highlights the use of deep learning for floor plan understanding, where CNN-based architectures are applied to segment and classify spatial components such as rooms, doors, and windows. However, most systems focus primarily on detection accuracy without integrating domain-specific evaluation frameworks. Although AI techniques such as YOLO and CNN-based models can effectively detect rooms and structural elements, many existing systems do not incorporate cultural and architectural principles such as Vastu compliance [4], [5], [6].

Vastu Shastra, an ancient Indian architectural science, provides guidelines for designing residential spaces based on directional alignment and energy flow. Despite its importance in many regions, Vastu evaluation is still largely manual and dependent on expert consultation. Existing approaches to Vastu compliance lack scalability and automation, making them unsuitable for modern digital applications.

Furthermore, traditional workflows in house planning are fragmented. As highlighted in the research document (page 1), tasks such as land analysis, floor plan design, interior optimization, and cultural compliance are handled independently, leading to inefficiencies and repeated iterations. Although some systems attempt to integrate optimization techniques such as genetic algorithms for layout design, they rarely combine these with AI-based detection and rule-based evaluation.

Therefore, there is a growing need for an intelligent system that integrates AI-based floor plan analysis with rule-based Vastu evaluation to provide automated, reliable, and user-friendly decision support.

1.1 Background & Literature Survey

The increasing integration of artificial intelligence into architectural and design-related applications has created new opportunities for automating traditionally manual evaluation processes. One such domain is Vastu Shastra, a traditional system of architectural principles that focuses on spatial arrangement, directional alignment, and energy flow within residential structures. In many regions, particularly in South Asia, Vastu compliance plays a significant role in decision-making during house design and property selection. However, the evaluation of Vastu compliance is typically performed manually by experts, making the process time-consuming, subjective, and often inconsistent.

The need for automation in spatial evaluation is not unique to Vastu analysis. In the broader architecture, engineering, and construction (AEC) domain, manual compliance checking has been identified as inefficient and prone to human error. Studies on automated compliance checking (ACC) systems have shown that translating design regulations into machine-readable formats can significantly improve consistency and reduce human effort [2], [3]. Although these systems are primarily designed for statutory building codes, the underlying concept of rule-based validation can be extended to traditional architectural principles such as Vastu.

In recent years, rule-based and knowledge-driven systems have been widely used to automate design validation tasks. Research in building information modeling (BIM) has demonstrated that structured rule representations and ontology-based reasoning can effectively interpret spatial layouts and validate them against predefined constraints [3], [9], [10]. These approaches are relevant to the present study, as Vastu evaluation also involves interpreting spatial relationships and applying rule-based logic to determine compliance.

A critical requirement for automating Vastu analysis is the ability to interpret floor plan images. Floor plans are complex visual documents that contain structural elements such as walls, doors, windows, and labeled rooms. Extracting meaningful information from such images has been a long-standing challenge in computer vision. Traditional approaches relied on handcrafted features and rule-based pipelines; however, these methods often struggled with variations in drawing styles and layouts.

The advancement of deep learning has significantly improved the performance of floor plan analysis systems. Convolutional Neural Networks (CNNs) have proven effective in extracting hierarchical features from images, enabling the detection and classification of complex objects [4], [5]. Recent research highlights that deep learning-based approaches outperform traditional methods in tasks such as room segmentation, symbol recognition, and object detection within floor plans [4]. This development provides a strong foundation for applying AI techniques to architectural image interpretation.

One notable study by Zeng et al. proposed a deep learning framework for floor plan recognition using multi-task neural networks with room-boundary-guided attention [5]. Their approach demonstrated that combining spatial context with visual features improves the accuracy of recognizing architectural elements. This finding supports the idea that floor plan understanding requires both local feature extraction and global spatial reasoning.

In addition to general floor plan analysis, some research efforts have explored the integration of computational methods with Vastu principles. A study on generating optimal residential layouts using Vaastu Shastra applied neural networks and fuzzy logic to evaluate compliance scores [1]. While this work demonstrates the feasibility of computational Vastu evaluation, it focuses on layout generation rather than analyzing existing floor plan images provided by users. Therefore, there remains a gap in developing systems that can interpret real-world floor plan images and apply Vastu rules dynamically.

Object detection has emerged as a key technique for interpreting visual data, particularly with the development of the YOLO (You Only Look Once) framework. YOLO models treat object detection as a single regression problem, allowing real-time detection with high efficiency [6]. Subsequent improvements in the YOLO family have enhanced detection accuracy and scalability, making it suitable for a wide range of applications [7]. These characteristics make YOLOv8 an appropriate choice for detecting multiple architectural elements within floor plan images.

Recent studies have also explored the application of YOLO-based models in detecting architectural components such as doors and windows. These studies demonstrate that modern object detection frameworks can effectively identify structural elements even in complex visual environments [8]. This is particularly relevant for Vastu analysis, where elements such as entrances, openings, and room placements are critical factors in evaluation.

Despite these advancements, existing research primarily focuses on either floor plan analysis or Vastu-based design generation, with limited integration between the two. There is a lack of systems that combine deep learning-based object detection, spatial reasoning, and rule-based Vastu evaluation into a unified, user-accessible platform. This gap highlights the need for a comprehensive solution that can automatically interpret floor plan images and provide meaningful Vastu compliance insights.

In summary, the literature indicates that while significant progress has been made in automated compliance checking, floor plan analysis, and object detection, there remains an opportunity to integrate these approaches into a unified system for Vastu evaluation. The present study addresses this gap by combining YOLOv8-based object detection, image processing techniques, spatial analysis, and rule-based reasoning to develop an AI-driven Vastu compliance evaluation system.

1.2. Research gap

Despite the significant progress in artificial intelligence, architectural analysis, and automated compliance checking, several limitations remain unaddressed in existing systems. These limitations create a clear research gap that motivates the development of the proposed AI-based Vastu detection and compliance evaluation system.

Firstly, most existing floor plan analysis systems focus primarily on **visual recognition tasks**, such as detecting walls, rooms, doors, and windows, without incorporating **domain-specific reasoning frameworks**. While deep learning models, particularly convolutional neural networks and object detection architectures, have achieved strong performance in recognizing spatial elements, they do not interpret the **functional or cultural significance** of these elements. In the context of Vastu Shastra, the placement of rooms relative to cardinal directions and spatial zones plays a critical role. However, current AI-based systems do not extend their capabilities beyond detection to evaluate such spatial relationships in a meaningful way.

Secondly, Vastu compliance evaluation remains largely **manual and expert-dependent**. Traditional Vastu analysis relies on practitioners who interpret floor plans based on directional alignment and spatial arrangement. This approach, while rooted in long-standing domain knowledge, introduces variability due to subjective interpretation. Similar challenges have already been identified in architectural compliance checking, where manual validation processes are considered time-consuming, inconsistent, and difficult to scale [2], [3]. Although automated compliance checking systems exist within the building industry, they are predominantly designed for **regulatory building codes** and often depend on structured formats such as Building Information Modeling (BIM). These systems do not address the evaluation of **rasterized floor plan images**, nor do they incorporate Vastu-specific principles.

Another critical gap lies in the **lack of integration between AI-based detection and rule-based reasoning systems**. Existing research typically treats these components independently. Deep learning models focus on feature extraction and classification, while rule-based systems operate on structured inputs. There is limited work that effectively combines these two paradigms into a unified pipeline capable of processing raw image data and producing interpretable compliance outputs. In the context of this project, such integration is essential because Vastu evaluation requires both accurate detection of spatial elements and logical reasoning based on predefined rules.

Furthermore, current floor plan analysis systems often lack **user interaction and adaptability**. Many existing tools operate as backend research models without providing user-friendly interfaces for real-world

applications. They do not allow users to define critical parameters such as **North orientation**, which is essential for directional analysis in Vastu evaluation. Without such inputs, the interpretation of spatial alignment becomes ambiguous and unreliable.

Another important limitation is the absence of **center-based spatial analysis**, particularly in relation to the concept of Brahmasthan. While some studies focus on detecting rooms and boundaries, they rarely compute the geometric center of the floor plan or evaluate whether this central zone remains unobstructed. This aspect is crucial in Vastu-based evaluation but is largely overlooked in existing automated systems.

Additionally, existing research often emphasizes **model performance metrics** such as precision, recall, and accuracy, without considering the broader objective of **decision support**. In practical applications, users require not only detection outputs but also meaningful insights, such as compliance scores, room-wise analysis, and actionable recommendations. Current systems do not sufficiently address this requirement.

Finally, there is a notable lack of **end-to-end systems** that integrate image preprocessing, object detection, spatial analysis, directional mapping, and rule-based evaluation into a single workflow. Most existing approaches focus on individual components rather than delivering a complete solution that can be deployed in real-world scenarios.

Based on these observations, the research gap can be summarized as follows:

There is no existing system that can **automatically analyze 2D floor plan images, detect architectural elements, determine their spatial orientation, and evaluate their compliance with Vastu principles in a unified, user-interactive environment**.

This gap directly motivates the development of the proposed system, which combines deep learning-based object detection with geometric analysis and rule-based reasoning to provide an automated and user-friendly Vastu compliance evaluation platform.

Reference	Functionalities of the Application	Technology and techniques Used for implementation	Platform	Limitations / Research Gap
Zeng et al. [5]	Deep floor plan recognition including walls, rooms, doors, and windows	Multi-task CNN with room-boundary-guided attention	Desktop / Research Models	No Vastu evaluation, no rule-based compliance checking
BIM-based ACC Systems [3], [9]	Automated checking of building compliance against regulations	Rule-based reasoning, ontology, BIM models and for segmentation	Desktop / Enterprise Systems	Works only with BIM data, not image-based floor plans, no Vastu integration
Vastu Layout Optimization System [1]	Generates optimized floor plans using Vastu principles	Neural networks + fuzzy logic scoring	Desktop / Simulation Tools	Does not analyze user-uploaded floor plans, no object detection
YOLO-based Architectural Detection [8]	Detection of doors and windows in architectural images	Human expertise and interpretation	Physical consultation	Time-consuming, subjective, no automation
Traditional Vastu Consultation	Manual evaluation of floor plans based on Vastu principles	Computer vision and transfer learning based on deep Convolutional Neural Networks (CNN)	Android mobile devices	Higher accuracy is expected
Proposed System (This Study)	Detects rooms, doors, windows and evaluates Vastu compliance with scoring and recommendations	YOLOv8, OpenCV, geometric analysis, rule-based evaluation engine	Web-based application	Addresses all above gaps through integrated AI + rule-based system

Table 1.2.1: Comparison of existing systems and identifying gaps

1.3. Research problem

The rapid growth of artificial intelligence and computer vision has enabled significant advancements in automated image analysis and object detection. However, despite these developments, there is still no practical system capable of interpreting 2D residential floor plan images and evaluating them against domain-specific spatial principles such as Vastu Shastra in an automated and user-interactive manner.

In real-world scenarios, homeowners and designers often rely on manual interpretation or expert consultation to assess whether a floor plan aligns with Vastu principles. This process is not only time-consuming but also subject to inconsistency due to variations in expert knowledge and interpretation. Furthermore, manual evaluation becomes increasingly difficult when dealing with complex floor layouts, where multiple spatial elements such as rooms, doors, windows, and staircases must be analyzed simultaneously in relation to directional orientation.

Although deep learning models, particularly object detection frameworks like YOLO, have demonstrated strong performance in identifying objects within images, they do not inherently provide contextual or rule-based reasoning. Similarly, rule-based systems require structured input data, which is not directly available from raw floor plan images. As a result, there exists a disconnect between visual detection capabilities and logical evaluation processes.

Another critical issue is the absence of systems that can accurately determine the **directional orientation of detected elements** based on a user-defined North reference. Direction plays a fundamental role in Vastu evaluation, yet most existing systems do not incorporate user-controlled orientation handling, leading to incomplete or incorrect spatial interpretation.

Additionally, key spatial concepts such as the **Brahmasthan (center of the house)** are not addressed in existing automated approaches. Determining whether this central zone is obstructed requires both geometric analysis and contextual understanding, which are not supported by current standalone detection models.

Moreover, existing approaches do not provide a **comprehensive evaluation output**, such as compliance scores, room-wise analysis, and actionable recommendations, which are essential for real-world usability. Users require interpretable results rather than raw detection outputs, yet this aspect is largely missing in current systems.

Therefore, the research problem addressed in this study can be stated as follows:

How to design and implement an automated system that can accurately analyze 2D floor plan images, detect architectural elements, determine their spatial orientation based on user-defined

direction, and evaluate their compliance with Vastu principles using a rule-based reasoning framework, while providing meaningful and interpretable outputs to end users.

1.4. Research Objectives

1.4.1. Main Objective

The primary objective of the proposed solution is to develop a web-based application that enables users to analyze residential floor plan images and evaluate their compliance with Vastu principles using artificial intelligence. The system is designed to accept a 2D floor plan image as input and automatically detect key architectural elements such as rooms, doors, windows, entrances, staircases, and other structural components using a deep learning-based object detection model.

The identification and interpretation process is achieved through a web-based interface where users can upload floor plan images and define the North orientation of the house. Based on this input, the system performs spatial analysis to determine the relative positions of detected elements with respect to cardinal directions.

Following the detection and spatial mapping, the system applies a rule-based Vastu evaluation engine to assess whether the arrangement of rooms and structural elements aligns with predefined Vastu principles. The output of the system includes a compliance score, detailed room-wise analysis, and recommendations to improve the Vastu alignment of the given floor plan.

The overall approach is intended to provide a user-friendly, automated alternative to traditional manual Vastu consultation, enabling consistent, scalable, and interpretable evaluation of residential layouts.

1.4.1. Specific Objectives

1. A structured dataset of residential floor plan images was to be prepared and utilized to train a deep learning-based object detection model capable of identifying architectural elements such as bedrooms, kitchens, toilets, entrances, staircases, doors, and windows.
 2. The collected dataset was to be processed using image preprocessing techniques, including contrast enhancement, noise reduction, and normalization, in order to improve detection accuracy when analyzing floor plan images.
- Since training a model entirely from scratch is computationally expensive and time-consuming, transfer learning was applied using a pre-trained YOLOv8 model to achieve efficient and accurate detection performance.

3. The most suitable deep learning model configuration was to be identified by experimenting with training parameters such as image resolution, batch size, and epochs, in order to obtain the best possible detection accuracy for floor plan elements.
- Based on experimental evaluation, the selected model was optimized to balance detection accuracy and computational efficiency for real-world usage.
4. A spatial analysis mechanism was to be developed to calculate the orientation of detected elements using user-defined North direction, enabling accurate mapping of rooms and structures to cardinal directions required for Vastu evaluation.
5. A geometric analysis approach was to be implemented to detect the outer boundary of the floor plan and determine the central region (Brahmasthan), allowing evaluation of whether this area is obstructed by structural elements.
6. A rule-based Vastu evaluation engine was to be designed and implemented to assess compliance based on predefined spatial rules, including room placement, directional alignment, entrance positioning, and structural constraints.
7. The trained detection model and evaluation logic were to be integrated into a web-based application using modern frontend and backend technologies, allowing users to upload floor plans and visualize results interactively.
- The system was designed to provide outputs such as detected object overlays, compliance scores, opening counts (doors and windows), Brahmasthan status, and Vastu recommendations in an intuitive and user-friendly interface.

2. METHODOLOGY

2.1 Understanding the key pillars of the research domain

The development of the proposed AI-Based Vastu Detection and Compliance Evaluation system relies on several core artificial intelligence and image processing concepts. These foundational pillars enable the system to accurately detect architectural elements from floor plan images and evaluate their compliance with Vastu principles.

2.1.1 Deep Learning

Deep learning is a subset of machine learning that utilizes multi-layered neural networks to automatically learn hierarchical representations of data. It has become a dominant approach for solving computer vision problems such as image classification, object detection, and segmentation [4], [5].

In the context of this system, deep learning is used to analyze 2D floor plan images and identify structural elements such as rooms, doors, and windows. Unlike traditional rule-based image processing methods, deep learning models can learn spatial patterns and visual features directly from annotated datasets, making them highly effective for interpreting complex floor plan layouts.

The YOLOv8 architecture was selected as the primary deep learning model due to its ability to perform efficient object detection. YOLO models treat object detection as a single regression problem, predicting bounding boxes and class probabilities simultaneously, which significantly reduces computational overhead compared to multi-stage detectors [6], [7].

This capability is particularly important for the proposed system, where quick and accurate detection is required to support real-time user interaction within a web application environment.

2.1.2 Convolutional Neural Network (CNN)

Convolutional Neural Networks (CNNs) are specialized deep learning architectures designed for processing grid-like data such as images. CNN-based methods are widely used in floor plan analysis because they can extract spatial features such as edges, shapes, textures, and structural patterns from visual data [4], [5].

In this system, CNNs form the backbone of the YOLOv8 model. The convolutional layers extract meaningful features from floor plan images, enabling the model to differentiate between different architectural components such as bedrooms, kitchens, toilets, and entrances.

Each convolutional layer applies multiple filters across the image, generating feature maps that highlight important patterns. These feature maps are then progressively refined through deeper layers, allowing the model to understand complex spatial relationships between objects.

The ability of CNNs to capture both low-level features (edges and lines) and high-level features (room structures and layouts) makes them highly suitable for floor plan analysis.

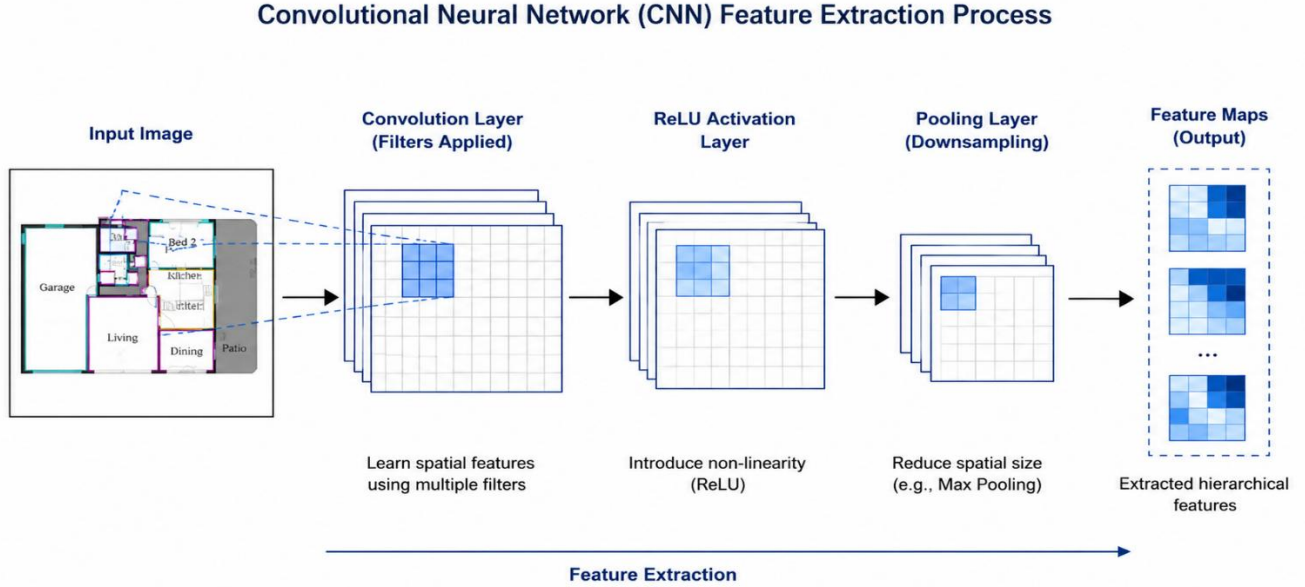


Figure 1 : CNN Feature Extraction Process showing how convolutional layers extract spatial features from input images.

2.1.3 Transfer Learning based on deep CNN

Training a deep learning model from scratch requires a large amount of labeled data and computational resources. However, the available dataset for this project consists of only 501 annotated images, which is relatively small for training a high-performance model.

To address this limitation, transfer learning was employed. Transfer learning involves using a pre-trained model that has already learned general image features from a large dataset and fine-tuning it on a specific task [4].

In this project, the YOLOv8 medium model (yolov8m.pt) pre-trained on the COCO dataset was used as the base model. The pre-trained weights allowed the model to leverage previously learned visual features such as edges, shapes, and object structures.

The model was then fine-tuned using the custom floor plan dataset to adapt it to the specific task of detecting architectural elements. This approach significantly reduced training time while improving

detection accuracy.

Training was performed using Google Colab with a Tesla T4 GPU. Although 150 epochs were initially requested, training was automatically stopped at epoch 54 due to early stopping, indicating convergence and preventing overfitting.

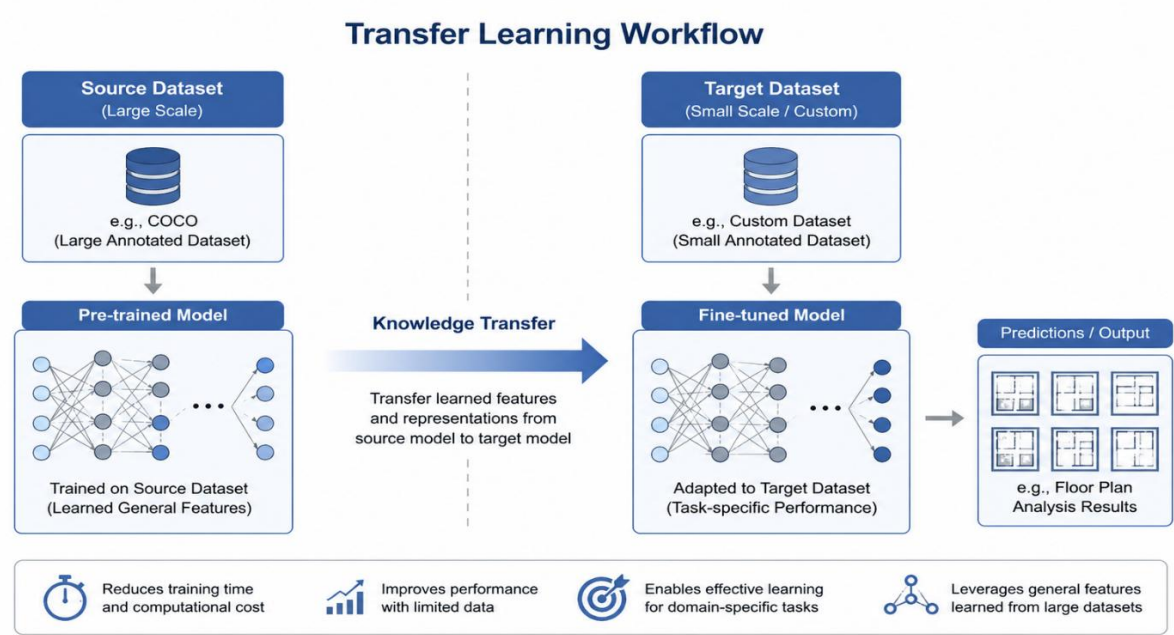


Figure 2: Transfer Learning Workflow showing knowledge transfer from a pre-trained model to a custom dataset.

2.1.4 Data Augmentation

Data augmentation is a technique used to artificially increase the size and diversity of a dataset by applying transformations to existing images. This helps improve the robustness and generalization ability of the model.

Given the limited dataset size, data augmentation played a crucial role in improving model performance. The following augmentation techniques were applied:

- Horizontal flipping
- Rotation between -10° and $+10^\circ$
- Brightness variation between -15% and $+15\%$

Additionally, preprocessing steps such as auto-orientation and resizing to 640×640 pixels were applied to standardize the dataset.

These transformations allowed the model to learn invariant features, ensuring that it performs well under different orientations and lighting conditions.

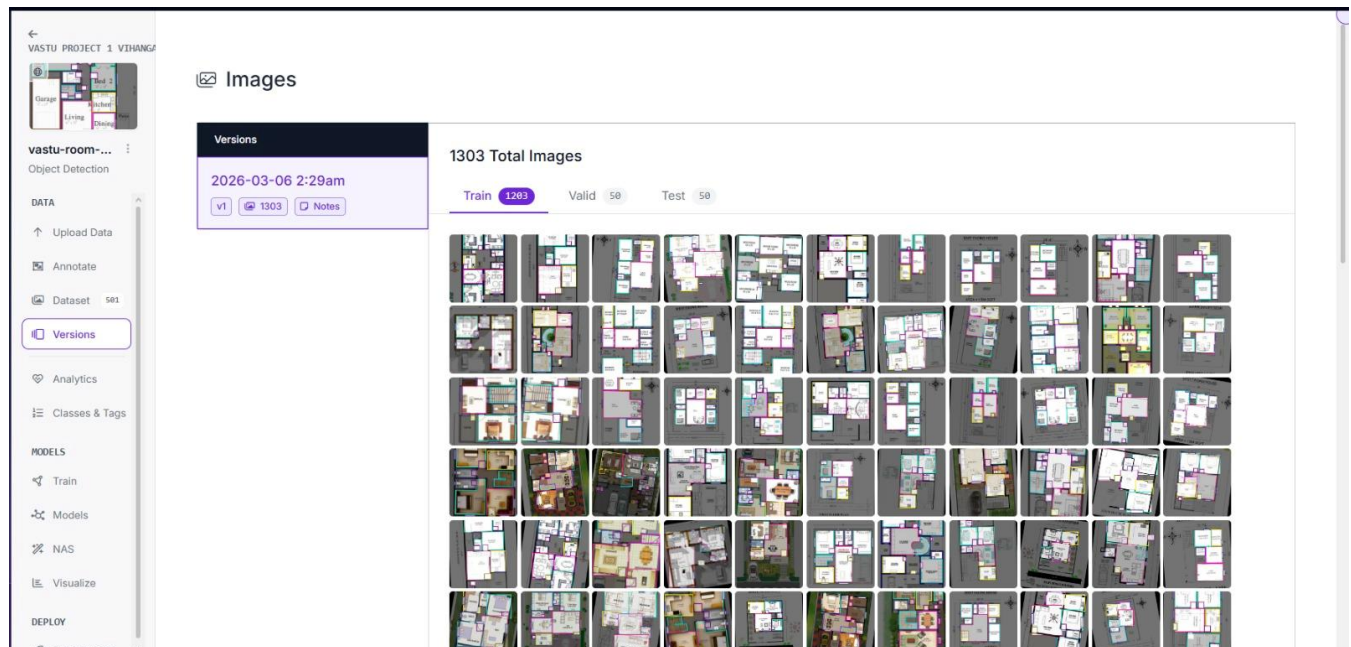


Figure 3: Augmentation Techniques applied to floor plan images including rotation, flipping, and brightness adjustments.

2.2 Approach

The approach of the proposed system integrates deep learning–based object detection with image processing techniques and a rule-based reasoning engine to perform automated Vastu compliance evaluation on 2D floor plan images. The overall workflow consists of data preparation, image preprocessing, object detection, spatial analysis, and rule-based evaluation.

2.2.1 Data Collection

The dataset used for training the object detection model was created and annotated using the Roboflow platform. A total of 501 2D residential floor plan images were collected, labeled, and prepared specifically for this study. The dataset includes annotations for 10 object classes, including bedrooms, kitchens, toilets, entrances, staircases, doors, and windows.

The dataset was divided into training, validation, and testing subsets, consisting of 401 training images, 50 validation images, and 50 testing images.

To ensure consistency, preprocessing steps such as auto-orientation and resizing to 640×640 pixels were applied. Additionally, data augmentation techniques including horizontal flipping, rotation between -10° and $+10^\circ$, and brightness variation between -15% and $+15\%$ were performed to enhance dataset diversity and improve model generalization.

The dataset was managed and exported using the Roboflow platform, which facilitated annotation, preprocessing, and augmentation.

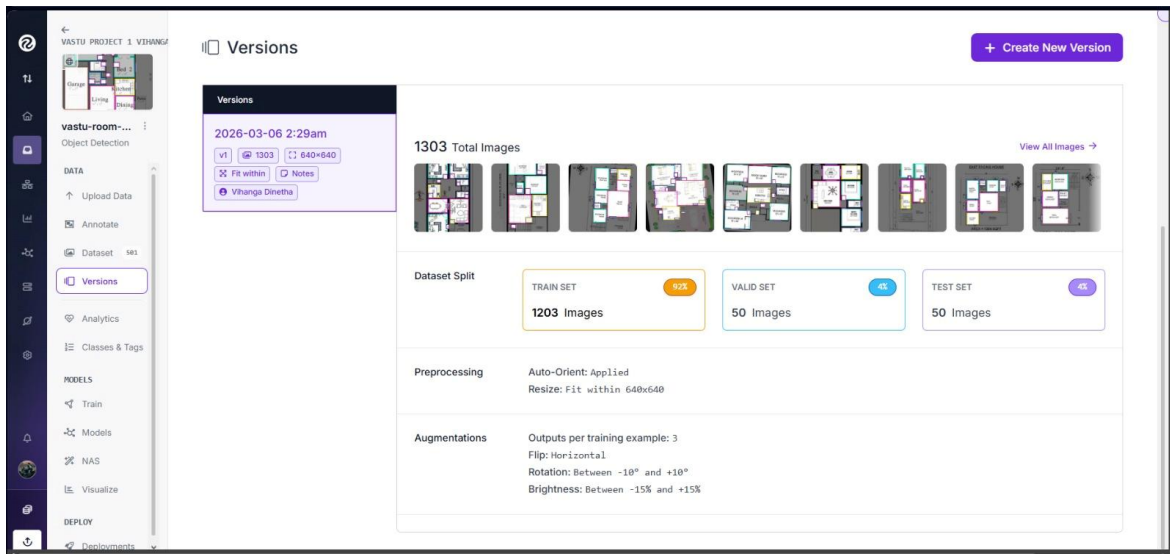


Figure 4: Dataset Distribution

2.2.2 Methodology

The methodology of the system follows a structured pipeline that transforms a raw floor plan image into a Vastu compliance evaluation result. This pipeline consists of multiple stages, each responsible for a specific task.

Image Input and Preprocessing

The process begins when the user uploads a 2D floor plan image through the web interface. Since floor plans may vary in quality, contrast, and noise levels, several preprocessing techniques are applied to enhance the image before detection.

Contrast Limited Adaptive Histogram Equalization (CLAHE) is used to improve local contrast and enhance important structural details. This is followed by bilateral filtering, which reduces noise while preserving edges. Canny edge detection is then applied to highlight structural boundaries such as walls and partitions.

Additionally, letterboxing is used to resize the image to the required input size (1024×1024) while preserving the aspect ratio, ensuring compatibility with the YOLOv8 model.

Object Detection using YOLOv8

The preprocessed image is passed to the YOLOv8 object detection model. The model predicts bounding boxes, class labels, and confidence scores for each detected object. The detection includes rooms (such as bedrooms, kitchens, and toilets) as well as structural elements like doors and windows.

To refine detection results, Non-Maximum Suppression (NMS) is applied to remove overlapping bounding boxes and retain the most confident predictions.

Boundary Detection and Brahmasthan Calculation

Following object detection, contour detection techniques are applied to identify the outer boundary of the house. This step is essential for determining the spatial layout of the floor plan.

Using the detected boundary, the geometric center of the house is calculated. This central region represents the Brahmasthan, which is a key concept in Vastu Shastra. The system evaluates whether this area is obstructed by any detected room or structure.

Direction Calculation

The direction of each detected object is calculated relative to a user-defined North direction. The system computes the angle between the object centroid and the center of the floor plan using the atan2 function.

Based on the calculated angle, each object is assigned to a directional category such as North, North-East, East, South-East, South, South-West, West, or North-West.

Vastu Rule-Based Evaluation

After spatial information is obtained, a rule-based engine evaluates the layout against predefined Vastu principles. These rules include conditions such as preferred room placements, restricted zones, and structural constraints.

Examples of applied rules include:

- Toilets should not be located in the North-East direction
- Kitchens are preferred in the South-East direction
- Brahmasthan should remain unobstructed
- Doors and windows are evaluated based on count parity

Each rule contributes to an overall compliance score. The system aggregates the results to generate a final Vastu compliance score, along with room-wise analysis and recommendations.

Output Generation

The final output is presented to the user through the web interface. This includes:

- A compliance score
- Visual overlay of detected objects
- Direction labels for each room
- Detailed explanation of rule violations
- Vastu recommendations

2.2.2.1 Transfer learning vs. Non-transfer learning

In this project, transfer learning was selected instead of training a model from scratch due to dataset size limitations and computational constraints.

Training a deep learning model from scratch requires a large-scale dataset and significant computational resources. With only 501 images available, a non-transfer learning approach would likely result in poor generalization and overfitting.

By using a pre-trained YOLOv8 model, the system benefits from previously learned features such as edge detection, object shapes, and spatial structures. These features are transferable to the floor plan domain, even though the original training dataset differs.

Fine-tuning the model on the custom dataset allows it to adapt to domain-specific features such as architectural layouts and room structures. This approach significantly improves performance while reducing training time.

The effectiveness of transfer learning is evident in the achieved performance metrics, where the model reaches a precision of 0.438 and recall of 0.455 despite the limited dataset size.

Transfer Learning (Fine-Tuning Pretrained Model)

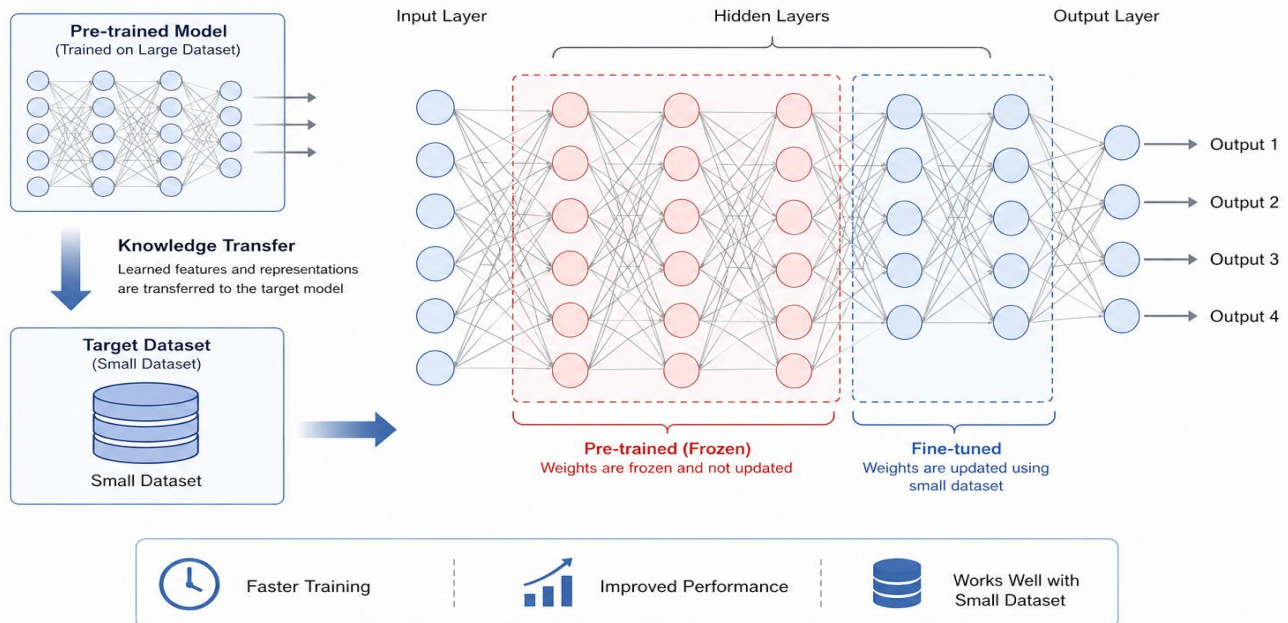


Figure 5: Transfer Learning approach where a pre-trained model is fine-tuned using a smaller dataset.

Non-Transfer Learning (Training from Scratch)

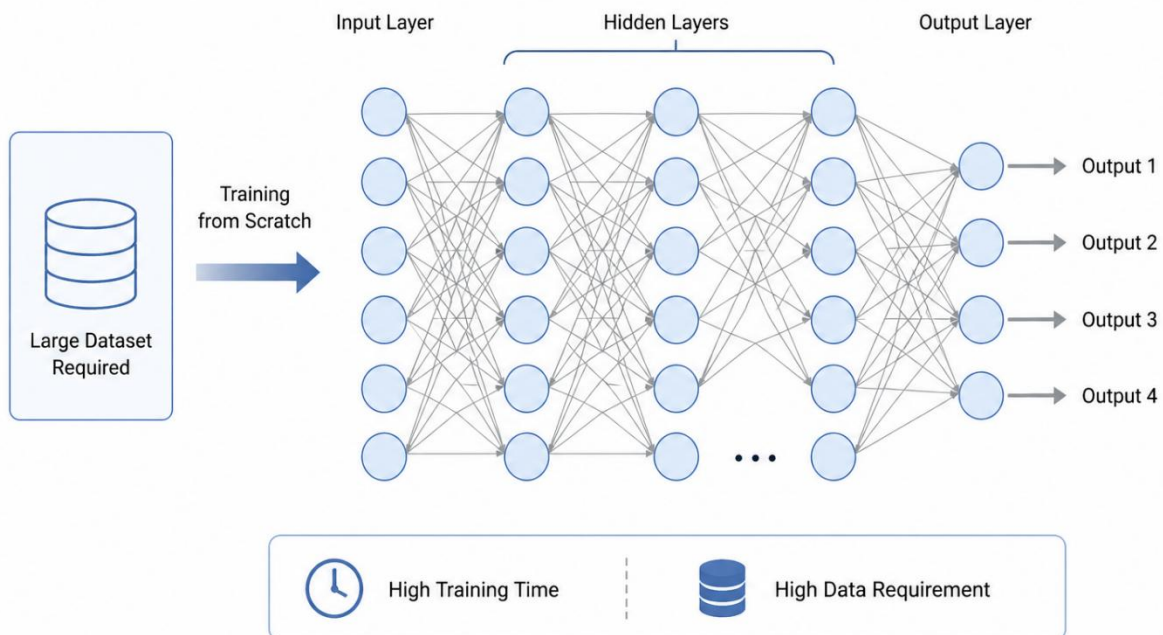


Figure 6: Non-Transfer Learning approach where the model is trained from scratch using a large dataset.

2.3 Summary of methodology

The methodology adopted for the proposed AI-Based Vastu Detection and Compliance Evaluation system integrates deep learning, image processing, and rule-based reasoning into a unified analytical pipeline. The system is designed to transform a raw 2D floor plan image into a structured Vastu compliance assessment through a sequence of well-defined computational stages.

At the initial stage, annotated floor plan data collected from Roboflow is used to train a YOLOv8-based object detection model. The dataset is preprocessed through resizing, auto-orientation, and augmentation techniques to improve generalization. Transfer learning is applied using the pre-trained yolov8m model to overcome the limitations of a relatively small dataset.

Once deployed, the system accepts a user-uploaded floor plan image and performs preprocessing operations including CLAHE for contrast enhancement, bilateral filtering for noise reduction, and Canny edge detection for structural boundary highlighting. These preprocessing steps ensure that key architectural features are preserved and enhanced prior to detection.

The preprocessed image is then passed to the YOLOv8 model, which detects rooms and structural components such as doors and windows. Non-Maximum Suppression is applied to refine overlapping detections and improve accuracy. Following detection, contour-based boundary extraction is used to identify the outer structure of the house, allowing the system to compute the geometric center known as the Brahmathan.

Subsequently, spatial analysis is performed by calculating the relative direction of each detected object using the atan2 function, based on a user-defined North reference. This enables classification of objects into cardinal and intercardinal directions.

Finally, a rule-based Vastu evaluation engine analyzes the spatial configuration against predefined Vastu principles. The system computes a compliance score, identifies violations, and generates recommendations. The results are presented through an interactive web interface, including visual overlays, directional labels, and explanatory feedback.

This integrated methodology ensures that the system effectively combines computer vision and traditional architectural principles to provide automated, consistent, and interpretable Vastu analysis.

Stage	Process	Methods / Algorithms Used	Output	Remarks
Data Preparation	Dataset collection and annotation	Roboflow dataset (501 images), augmentation techniques	Structured labeled dataset	Limited dataset size handled via augmentation
Image Preprocessing	Enhance image quality	CLAHE, Bilateral Filtering, Canny Edge Detection, Letterboxing	Enhanced input image	Improves detection accuracy
Object Detection	Detect rooms and structural elements	YOLOv8 (Ultralytics), CNN backbone, NMS	Bounding boxes with class labels	Real-time detection capability
Boundary Detection	Identify house layout	Contour Detection (OpenCV)	House boundary and center point	Required for Brahmasthan calculation
Direction Analysis	Assign directional zones	atan2 angle calculation	Directional labels (N, NE, E, etc.)	Depends on user-defined North
Vastu Evaluation	Apply compliance rules	Rule-based reasoning engine	Compliance score and violations	Based on traditional Vastu rules
Output Generation	Display results to user	React UI, Konva overlay, Framer Motion	Visual + textual feedback	User-friendly interpretation

Table 2.3.1: Summary of Methodology for AI-Based Vastu Detection and Compliance Evaluation System

The summarized methodology highlights the integration of multiple computational techniques, combining deep learning-based detection with spatial reasoning and rule-based evaluation. This multi-stage design ensures both accuracy in object detection and interpretability in Vastu compliance analysis.

2.4 High-Level Architecture Diagram

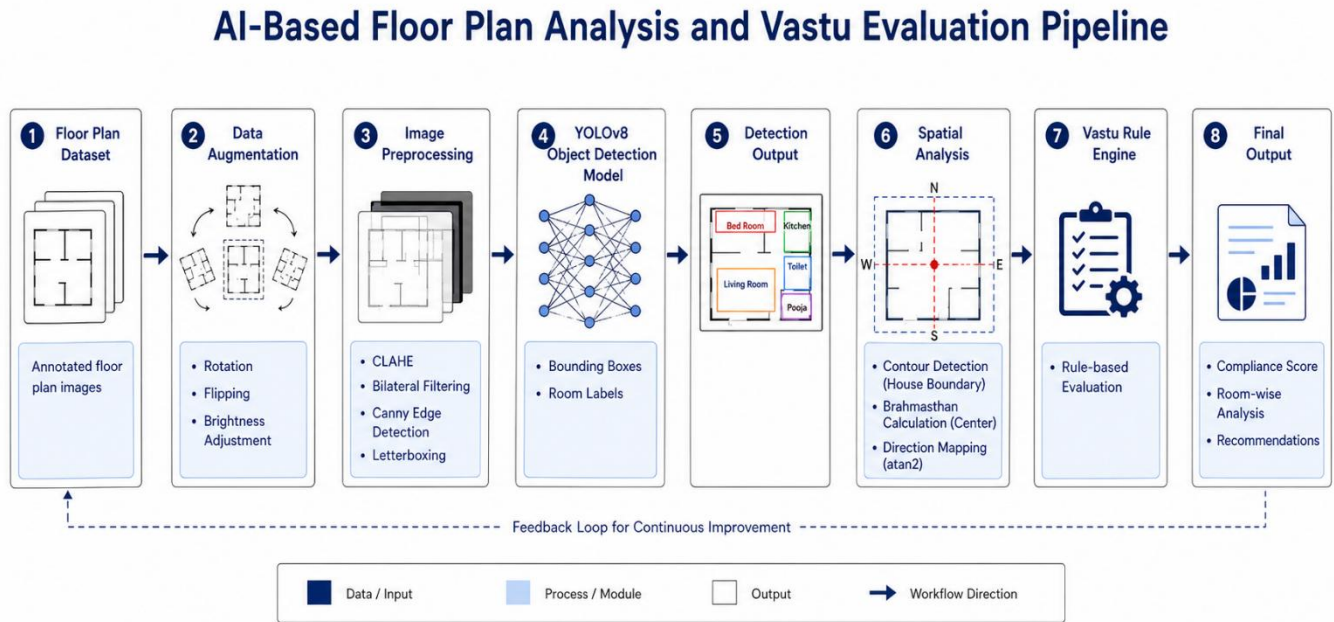


Figure 7: AI Detection and Vastu Evaluation Pipeline showing the transformation from floor plan input to compliance output

2.5 Project Requirements that have been achieved

2.5.1. Functional Requirements

1. The system shall allow users to upload a 2D floor plan image in standard formats such as JPEG or PNG through the web interface.
2. The system shall allow the user to define the North direction manually to ensure accurate directional mapping.
3. The system shall preprocess the uploaded image using techniques such as CLAHE, bilateral filtering, and edge detection to improve detection quality.
4. The system shall detect architectural elements including rooms, doors, and windows using the YOLOv8 object detection model.
5. The system shall display detected elements with bounding boxes and labels overlaid on the floor plan image.
6. The system shall calculate the spatial direction of each detected element relative to the user-defined North using angle-based computation.
7. The system shall identify the Brahmadhan (center of the house) using contour detection and

geometric center calculation.

8. The system shall determine whether the Brahmasthan is obstructed by any detected structure.
9. The system shall apply a rule-based Vastu evaluation engine to analyze the detected layout.
10. The system shall generate a compliance score based on rule satisfaction.
11. The system shall provide room-wise analysis indicating whether each detected space follows Vastu principles.
12. The system shall generate Vastu recommendations and tips based on detected violations.
13. The system shall allow users to manually configure or adjust room types and directions when automatic detection is inaccurate.
14. The system shall display analysis results, including compliance score, rule violations, and suggestions, in a structured and user-friendly interface.

2.5.1 Non-Functional Requirements

1. The system shall provide fast response time, completing detection and analysis within a few seconds.
2. The system shall maintain acceptable accuracy levels for object detection and spatial analysis based on trained YOLOv8 performance metrics.
3. The system shall be reliable and capable of handling different types of floor plan images with varying quality and complexity.
4. The system shall be user-friendly, allowing non-technical users to easily upload images, define directions, and understand analysis results.
5. The system shall provide clear visual feedback, including bounding boxes, labels, and directional indicators.
6. The system shall be scalable, allowing future expansion such as additional room categories, improved datasets, and extended Vastu rules.
7. The system shall be maintainable through modular architecture, enabling independent updates of AI models, preprocessing, and rule engines.
8. The system shall be compatible with modern web browsers and operate efficiently as a web-based application.
9. The system shall ensure security by implementing authentication mechanisms using Firebase Authentication.
10. The system shall handle user-uploaded data safely without unauthorized access or misuse.

2.5.3 Other Requirements

The system requires a stable internet connection to enable users to upload floor plan images and perform server-side analysis efficiently. Additionally, the application is designed to operate on modern web browsers such as Google Chrome, Microsoft Edge, or Mozilla Firefox to ensure compatibility and optimal performance. The backend server must be continuously available to handle image processing, object detection, and Vastu evaluation tasks. Furthermore, sufficient computational resources are required on the server side to execute the YOLOv8 model effectively and maintain acceptable response times during analysis.

2.6. Consideration of the aspects of the system

The design and implementation of the AI-Based Vastu Detection and Compliance Evaluation system are influenced by several practical, technical, and operational considerations. These considerations ensure that the system remains realistic, reliable, and applicable within its intended scope. They include assumptions made during development, system limitations, constraints, and potential risks that may affect overall performance.

2.6.1 Social aspects

The proposed system is designed to assist users in evaluating residential floor plans according to Vastu principles, which are culturally significant in many regions. Therefore, the system has a strong social relevance, particularly for individuals who seek guidance in house planning but may not have access to professional Vastu consultants.

By providing automated analysis, the system democratizes access to Vastu evaluation, allowing users to make informed decisions during house design or property selection. This reduces dependency on expert consultation and makes traditional knowledge more accessible to a wider audience.

However, it is important to recognize that Vastu Shastra is interpretative and may vary across practitioners and regions. Therefore, the system must be understood as a supportive tool rather than a definitive authority. Users should be aware that the recommendations generated are based on predefined computational rules and may not fully capture all traditional interpretations.

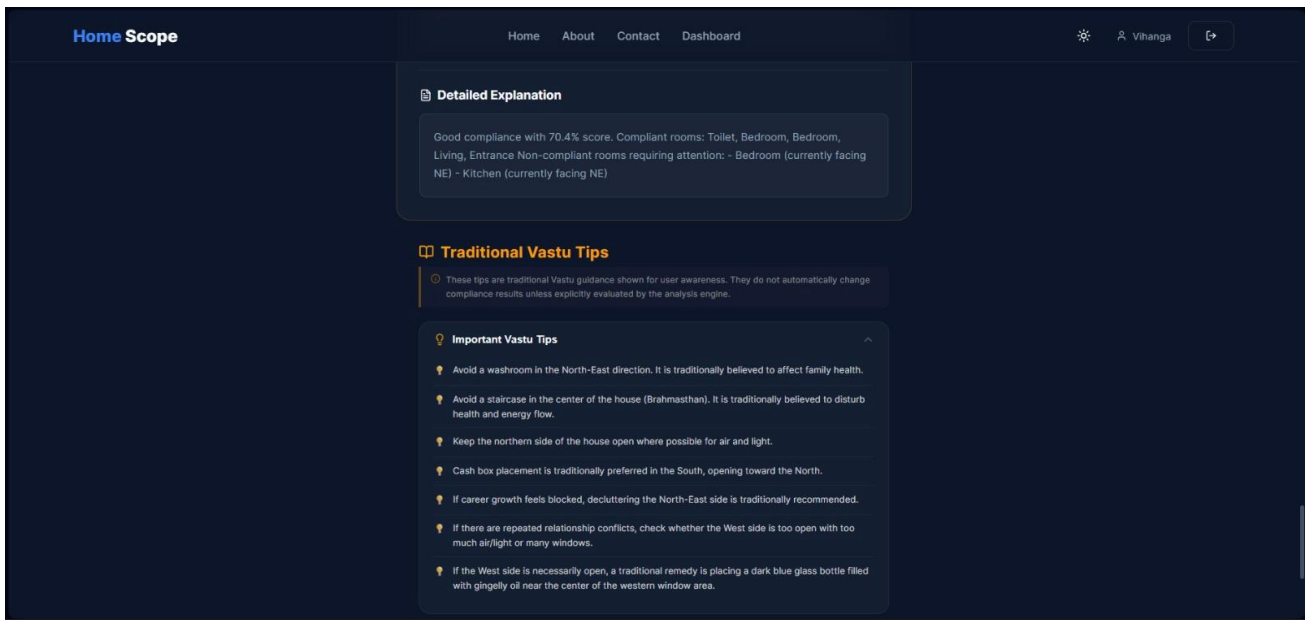


Figure 8: Vastu tips UI Section 1

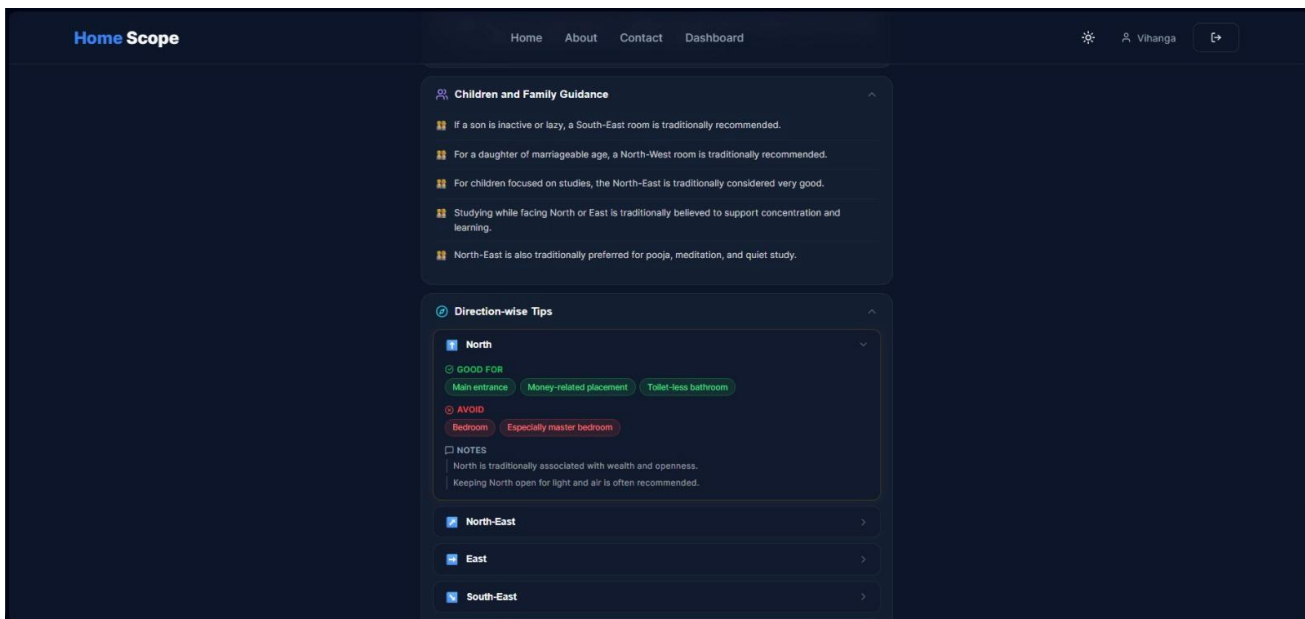


Figure 9: Vastu tips UI Section 2

2.6.2 Security aspects

Security is an important consideration in the system, particularly in handling user data and uploaded floor plan images. The application uses Firebase Authentication to ensure that only authorized users can access system features, thereby preventing unauthorized usage.

All interactions between the frontend and backend are handled through API calls, which should be secured to prevent misuse. Although the system does not store sensitive personal information, uploaded floor plan images may still be considered private data. Therefore, it is essential that these images are processed securely and not retained unnecessarily after analysis.

In addition, backend endpoints must be protected against common vulnerabilities such as unauthorized access, injection attacks, and excessive request loads. Proper validation of user inputs and rate limiting mechanisms can help ensure system stability and security.

2.6.3 Ethical aspects

The ethical considerations of the system mainly relate to transparency, reliability, and responsible use of AI-generated results. Since the system provides Vastu compliance scores and recommendations, there is a risk that users may interpret these outputs as absolute truths.

To address this, the system must clearly communicate that the results are generated based on algorithmic interpretation and predefined rules. It should avoid making deterministic claims that could mislead users into making critical decisions solely based on system output.

Additionally, fairness must be considered in AI predictions. The YOLOv8 model is trained on a limited dataset of 501 images, which may not represent all possible architectural styles. This introduces potential bias in detection performance, especially for uncommon layouts.

Therefore, ethical deployment requires transparency about model limitations and encouragement for users to use the system as a guidance tool rather than a final decision-maker.

2.6.4 Limitations

- The object detection model used in the system demonstrates moderate performance, with a precision of 0.438 and recall of 0.455. As a result, certain elements in the floor plan may not be detected accurately.
- Some object classes such as toilets, staircases, and master bedrooms show weaker detection performance, which may lead to incorrect classification or missed detections.
- The system is highly dependent on the quality of the input floor plan image. Low-resolution, blurred, or highly complex images can significantly reduce detection accuracy.
- Variations in floor plan styles, symbols, and drawing conventions may affect the consistency and reliability of object detection results.
- The accuracy of directional analysis depends entirely on the user-defined North direction. Incorrect selection of orientation can lead to completely inaccurate Vastu evaluation results.
- The Brahmasthan (center of the house) is calculated using geometric center estimation based on contour detection. This approach may not be fully accurate for irregular or complex floor layouts.
- The Vastu evaluation is based on a predefined rule-based system. Therefore, it does not cover all possible Vastu principles and lacks adaptability to different interpretations without manual updates.
- The dataset used for training consists of 501 images, which may not represent all real-world floor plan variations, potentially affecting the generalization capability of the model.

2.7. Commercialization aspects of the product

2.7.1 Target Audience

- Homeowners and individuals who are planning to build or purchase houses and require Vastu compliance evaluation before making decisions
- Architects and civil engineers who need to validate floor plan designs according to Vastu principles during the design phases
- Real estate developers and property agents who want to provide Vastu-based analysis as a value-added service to attract customers
- Interior designers and consultants who consider spatial arrangements and directional placements in residential planning
- General users who are interested in understanding Vastu principles and analyzing their existing house layouts.

2.7.2 Market Space

- No strict age limitation for users, as the system is designed to be simple and user-friendly
- No requirement for advanced technical knowledge, allowing non-technical users to easily interact with the system
- No prior knowledge of Vastu Shastra is required, as the system provides automated analysis and clear explanations
- Applicable in regions where Vastu principles are culturally significant and widely practiced
- Can be extended as a web-based Software-as-a-Service (SaaS) platform accessible globally

2.7.3 Revenue Earning

- Revenue can be generated through subscription-based models for accessing advanced Vastu analysis features
- Additional income can be earned through premium services such as detailed reports, enhanced recommendations, and 3D visualization features
- Enterprise-level solutions can be provided for architects, developers, and real estate companies with bulk analysis capabilities

- API-based integration services can be offered to external platforms such as property listing websites
- Revenue opportunities can also be explored through partnerships, advertisements, and collaborations with industry professionals

2.8 Testing and Implementation

The testing and implementation phase ensures that the developed system functions correctly, efficiently, and reliably under different scenarios. This section describes how the AI-based Vastu detection system was implemented and how its performance was validated through systematic testing.

2.8.1 Code Implementations for the research part

The implementation of the proposed system is based on a combination of frontend, backend, and AI components that work together to perform floor plan analysis and Vastu evaluation.

The frontend of the system was developed using React 19 and Vite, providing an interactive user interface for uploading floor plan images, selecting the North direction, and visualizing analysis results. Konva was used to render bounding boxes and detected elements on the 2D canvas, while Framer Motion was used to enhance user experience through smooth animations. Firebase Authentication was integrated to manage user access securely.

The backend was implemented using Python 3 with FastAPI and Uvicorn. The backend is responsible for handling API requests, processing uploaded images, performing AI inference, and returning structured results to the frontend.

The AI component of the system is based on the YOLOv8 object detection model. The model was trained using a Roboflow dataset consisting of 501 images, including 401 training images, 50 validation images, and 50 testing images. The training process was carried out in the Google Colab environment using a Tesla T4 GPU. The pretrained model yolov8m.pt was fine-tuned for the task of detecting architectural elements such as rooms, doors, and windows.

Before passing images to the detection model, several preprocessing techniques were applied. These include CLAHE for contrast enhancement, bilateral filtering for noise reduction, and Canny edge detection to improve boundary clarity. Letterboxing was used to resize images to the required input dimensions without distortion.

After detection, Non-Maximum Suppression (NMS) was applied to remove duplicate bounding boxes and improve prediction quality. The detected elements were then processed to calculate spatial positions and directions using the atan2 function based on the user-defined North direction.

Contour detection was used to estimate the outer boundary of the house, enabling the calculation of the Brahmasthan (central region). Based on the detected elements and their positions, a rule-based Vastu evaluation engine was applied to generate compliance scores, room-wise analysis, and recommendations.

2.8.2 Testing

Testing was carried out to evaluate both the functional correctness and performance of the system. The testing process included model evaluation, system-level testing, and user interface validation.

The performance of the YOLOv8 model was evaluated using standard object detection metrics. The model achieved a precision of 0.438, recall of 0.455, mAP@50 of 0.354, and mAP@50–95 of 0.268. These results indicate moderate detection capability, with strong performance in detecting doors, entrances, and windows, while certain classes such as toilets and staircases showed weaker performance.

System testing was conducted by uploading various floor plan images with different layouts, orientations, and complexities. The system was verified to correctly detect elements, calculate directions, and generate Vastu compliance results. Special attention was given to validating the correctness of directional mapping and Brahmasthan detection.

User interface testing was performed to ensure that all components, including image upload, North selection, room configuration, and result display, functioned as expected. The responsiveness and usability of the interface were also evaluated.

Additionally, edge case testing was conducted to identify system behavior under challenging conditions such as low-quality images, missing elements, or incorrect user inputs. These tests helped identify limitations and ensure system robustness.

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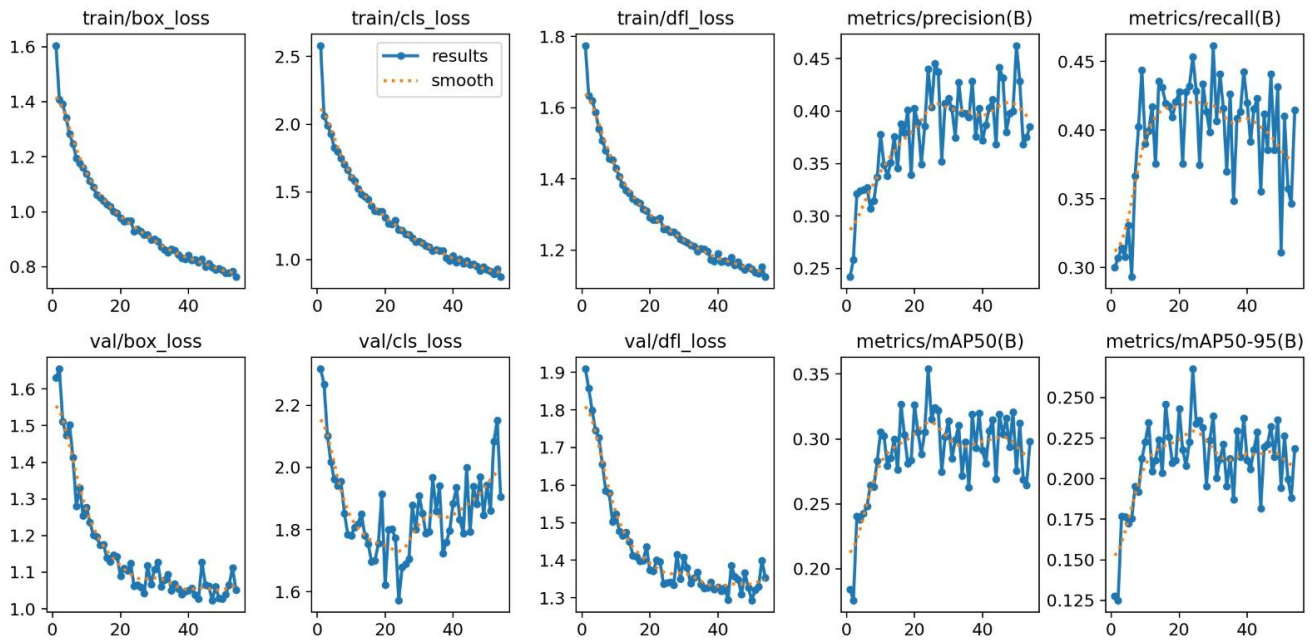


Figure 10: Model Performance Graphs

2.9 Tools and Technologies

1. Python 3
2. FastAPI
3. Uvicorn
4. YOLOv8
5. OpenCV
6. Google Colab
7. React + Vite
8. Konva
9. React Three Fibre
10. Framer Motion
11. Firebase Authentication

Research Area – Deep Learning Algorithms (object detection using YOLOv8), Image Processing Techniques, and Rule-Based Spatial Analysis for Vastu Evaluation.

3. RESULTS AND DISCUSSION

3.1 Results

The results of the proposed AI-Based Vastu Detection and Compliance Evaluation system are evaluated based on the performance of the YOLOv8 object detection model and the overall system functionality. The evaluation includes quantitative metrics obtained during model training as well as qualitative observations from system-level testing.

The YOLOv8 model was trained using a dataset consisting of 501 floor plan images, with 401 images used for training, 50 for validation, and 50 for testing. The model training process was conducted using a Tesla T4 GPU in the Google Colab environment, and training was stopped at epoch 54 due to early stopping.

The performance of the model was evaluated using standard object detection metrics, including precision, recall, mean Average Precision at IoU threshold 0.50 (mAP@50), and mean Average Precision across IoU thresholds from 0.50 to 0.95 (mAP@50–95).

The final evaluation metrics obtained are as follows:

- Precision: 0.438
- Recall: 0.455
- mAP@50: 0.354
- mAP@50–95: 0.268

These values indicate that the model has a moderate ability to correctly detect and classify objects within floor plan images. The recall value suggests that the model is able to identify a reasonable proportion of relevant objects, while the precision value indicates that some false detections are present.

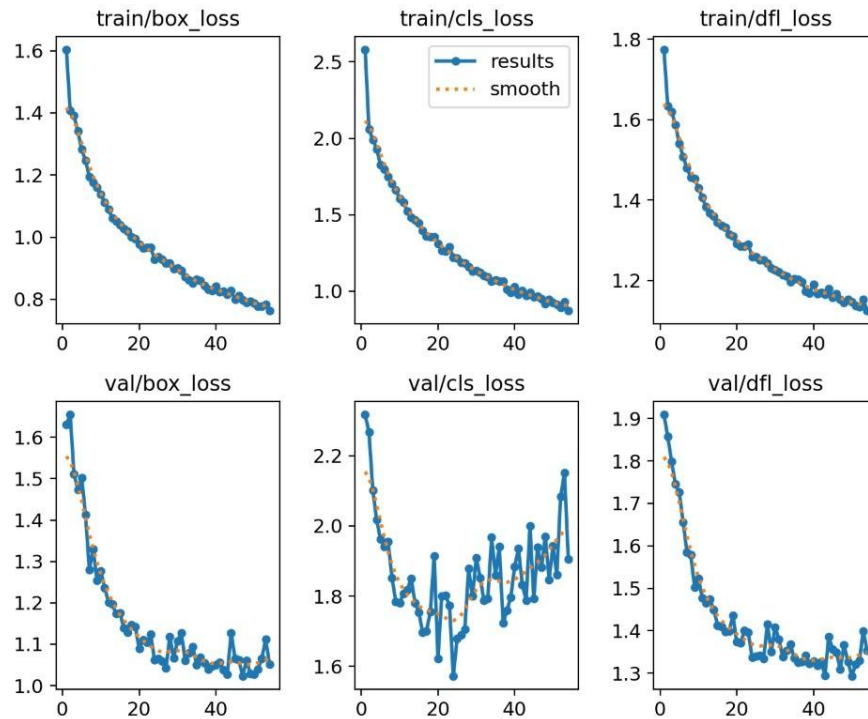


Figure 11: Training and Validation Loss Graphs

- Loss decreases → model is learning
- Validation loss stabilizes → good generalization

The training and validation loss graphs demonstrate a consistent decrease in loss values over the training epochs. This indicates that the model successfully learned relevant features from the dataset. The validation loss curves show a gradual stabilization, suggesting that the model does not severely overfit during training.

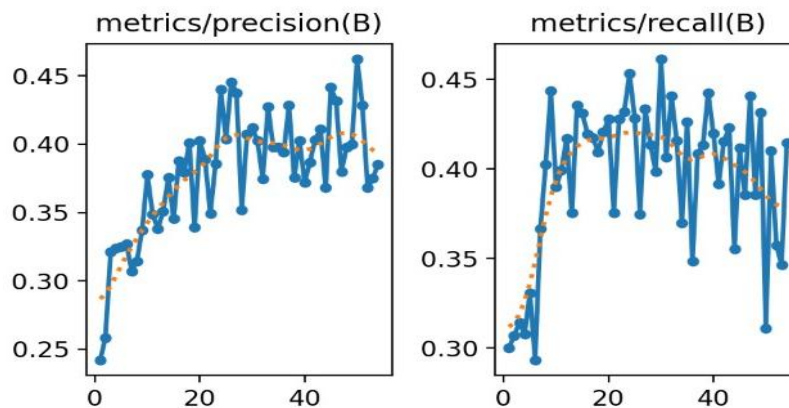


Figure 12: Precision and Recall Graphs

The precision and recall graphs show gradual improvement during the initial training stages, followed by

fluctuations. This behavior is expected due to the limited size of the dataset and variations in floor plan complexity. The recall remains slightly higher than precision, indicating that the model is more inclined to detect objects even at the cost of some false positives.

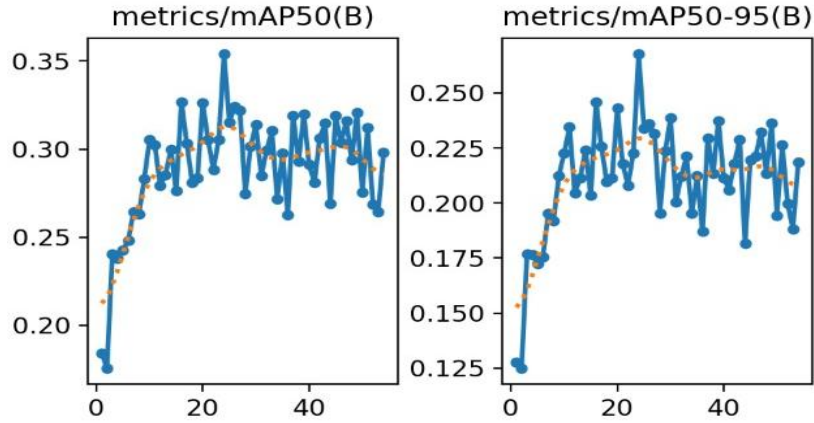


Figure 13: mAP Performance Graphs

The mAP@50 and mAP@50–95 graphs indicate moderate detection performance across object classes. The relatively lower mAP@50–95 value reflects the difficulty of achieving high localization accuracy across multiple IoU thresholds, especially in complex floor plan images.

From a class-wise perspective, the model demonstrates strong performance in detecting certain categories such as living rooms, doors, entrances, and windows. These elements typically have more consistent visual patterns and clearer boundaries in floor plans, making them easier to detect. In contrast, classes such as toilets, staircases, and master bedrooms show weaker performance. This may be due to limited training samples, variation in labeling, and differences in representation across floor plan images.

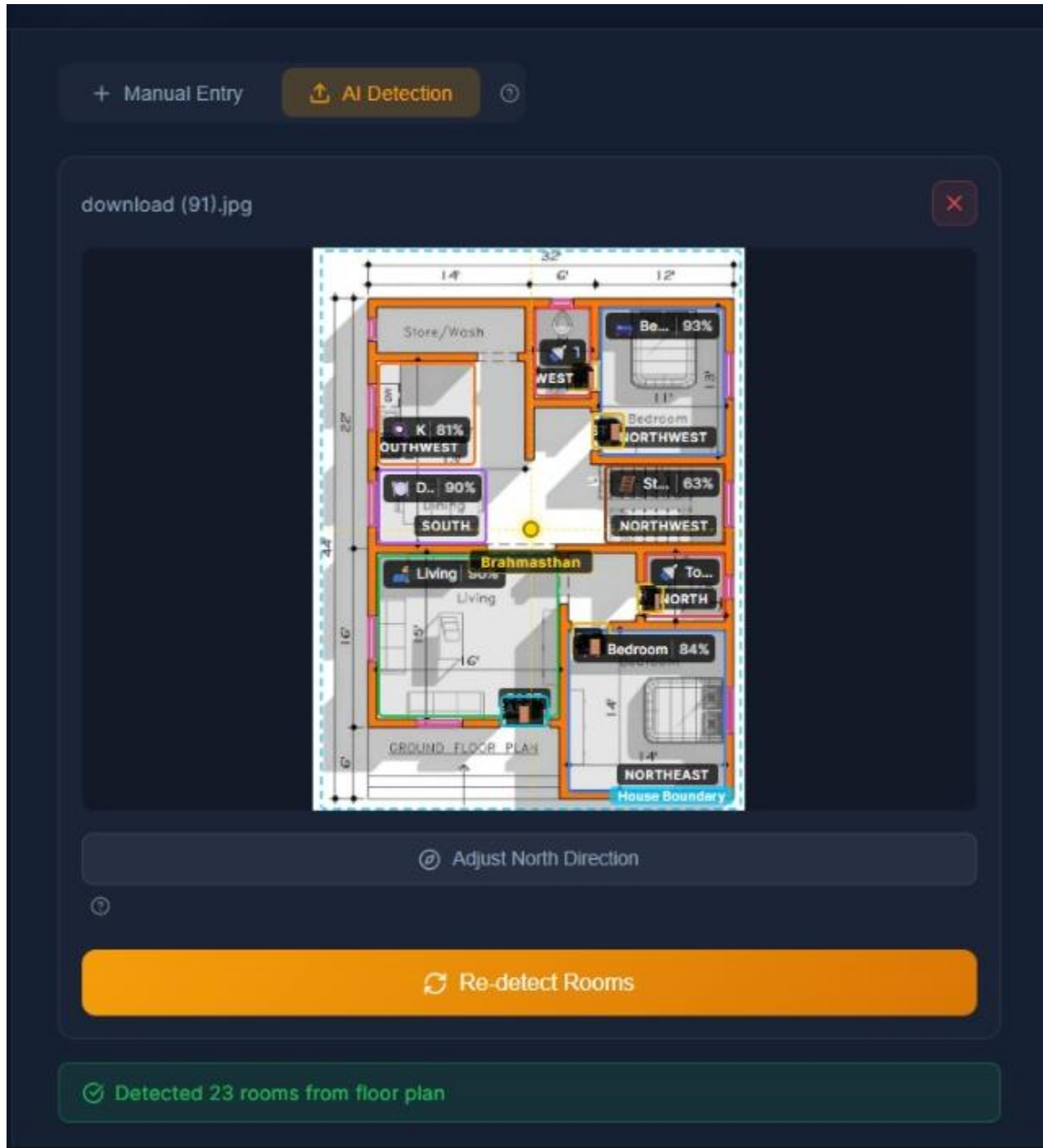


Figure 14: Detection Output

In addition to model-level evaluation, system-level results were also analyzed. The application successfully processes user-uploaded floor plan images, detects architectural elements, calculates directional orientation based on user-defined North, and generates Vastu compliance results. The system produces a compliance score along with room-wise analysis and recommendations. The Brahmasthan is identified and visually represented, and the system correctly evaluates whether it is occupied or free.

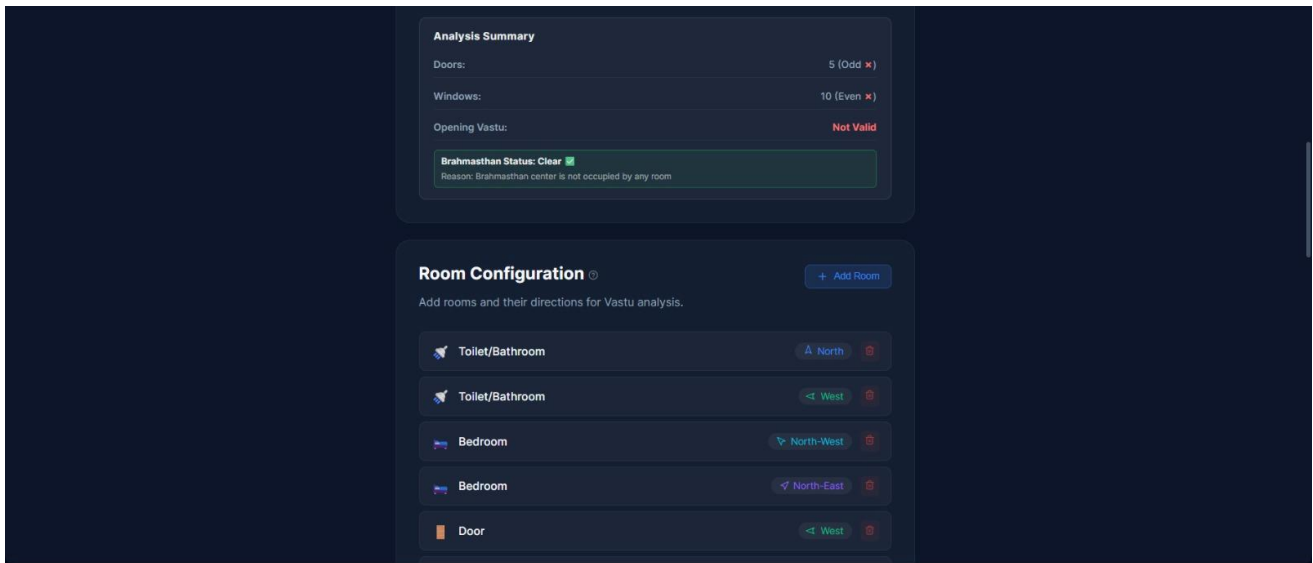


Figure 15: Room Configuration UI 1

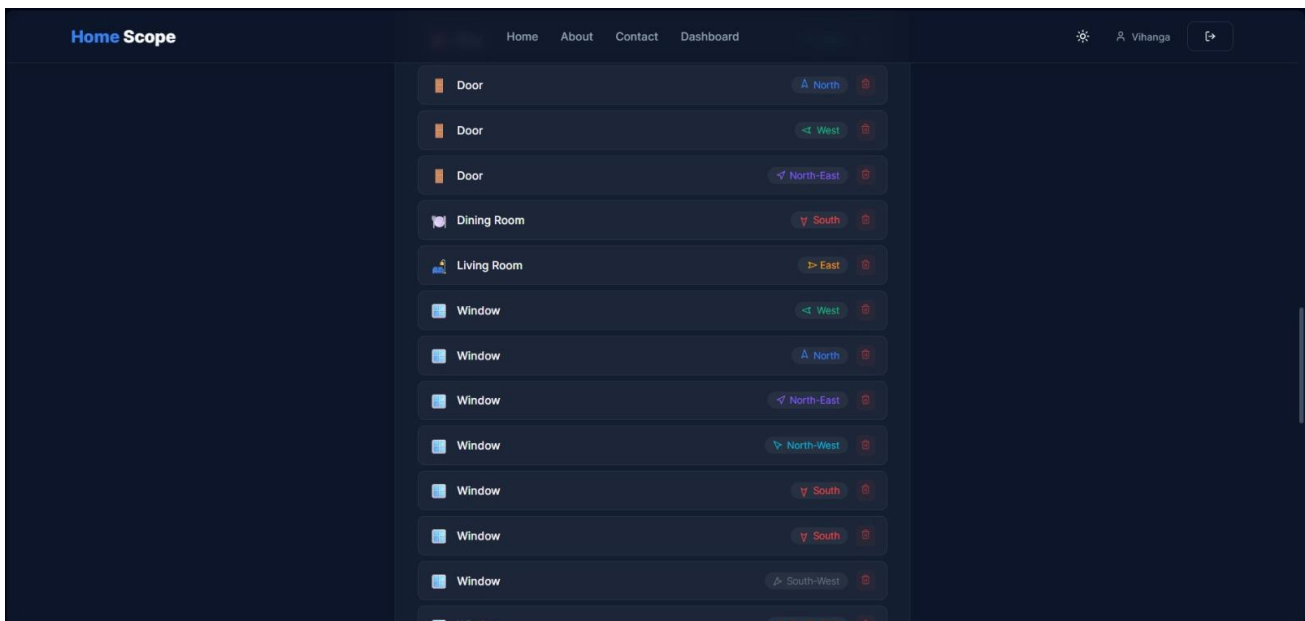


Figure 16: Room Configuration UI 2

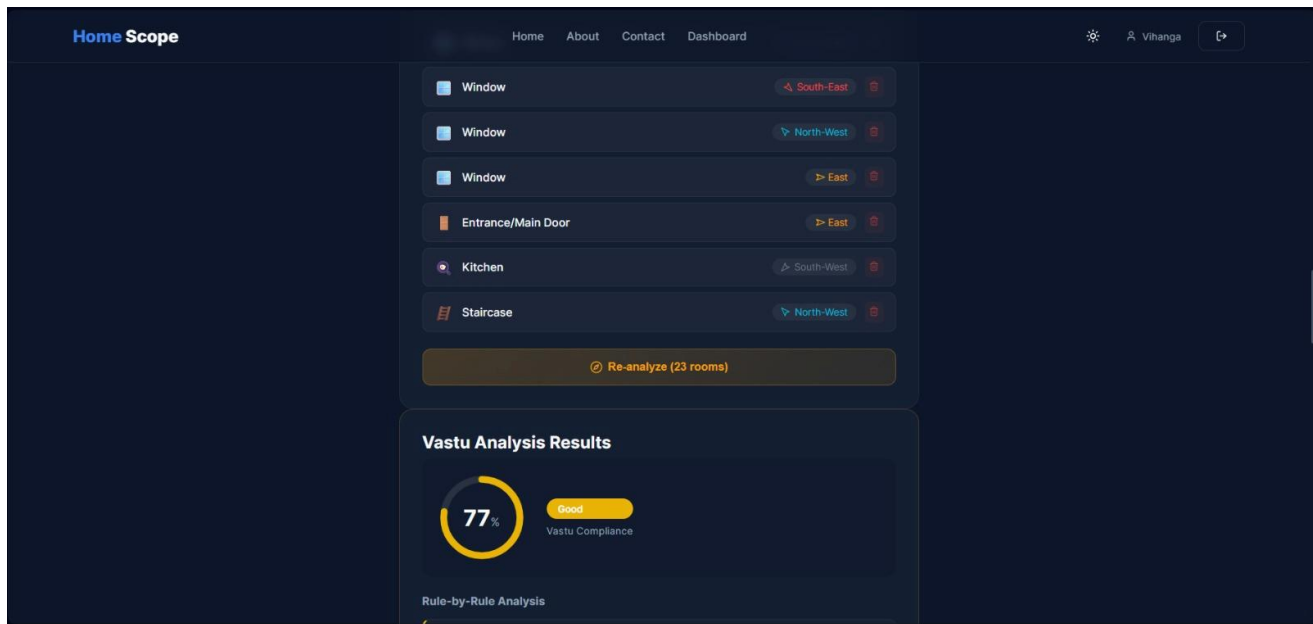


Figure 17: Vastu Compliance Score UI

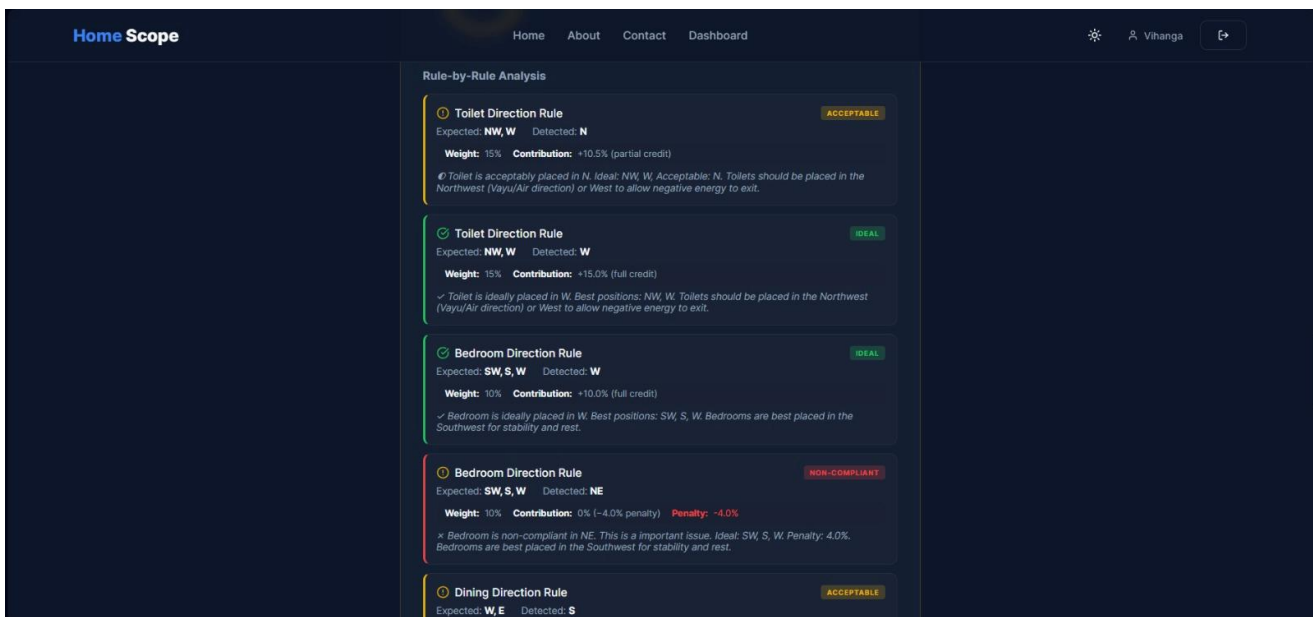


Figure 18: Vastu Rule-by-Rule Analysis UI I

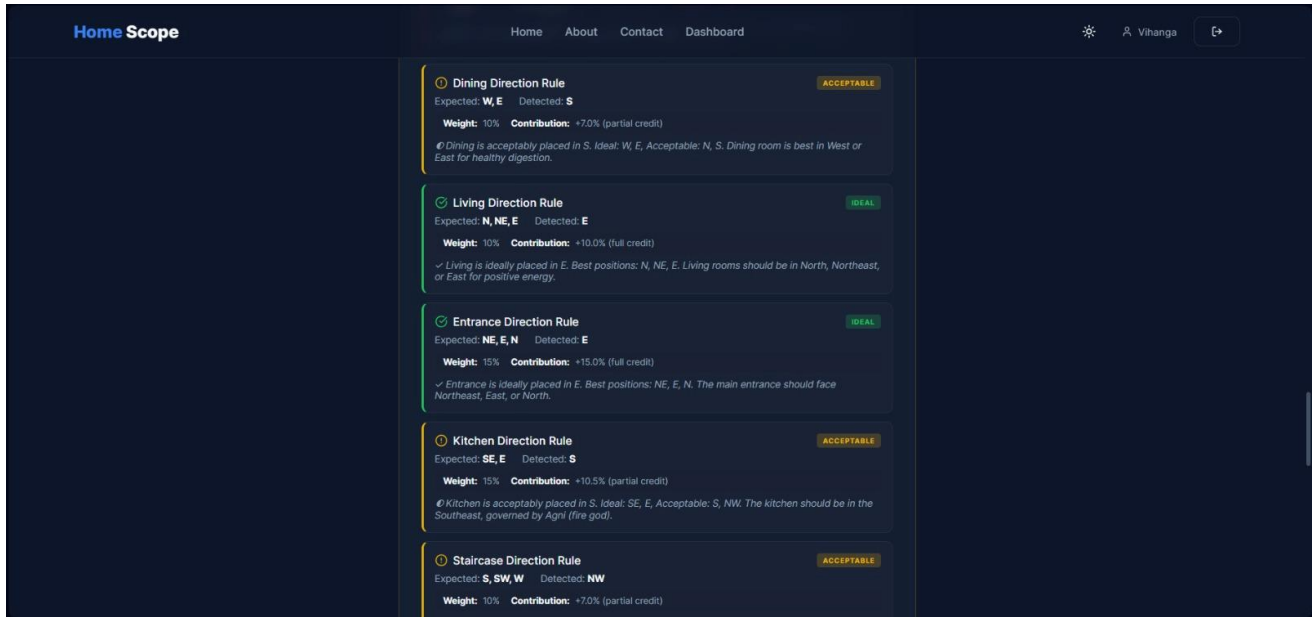


Figure 19: Vastu Rule-by-Rule Analysis UI 2

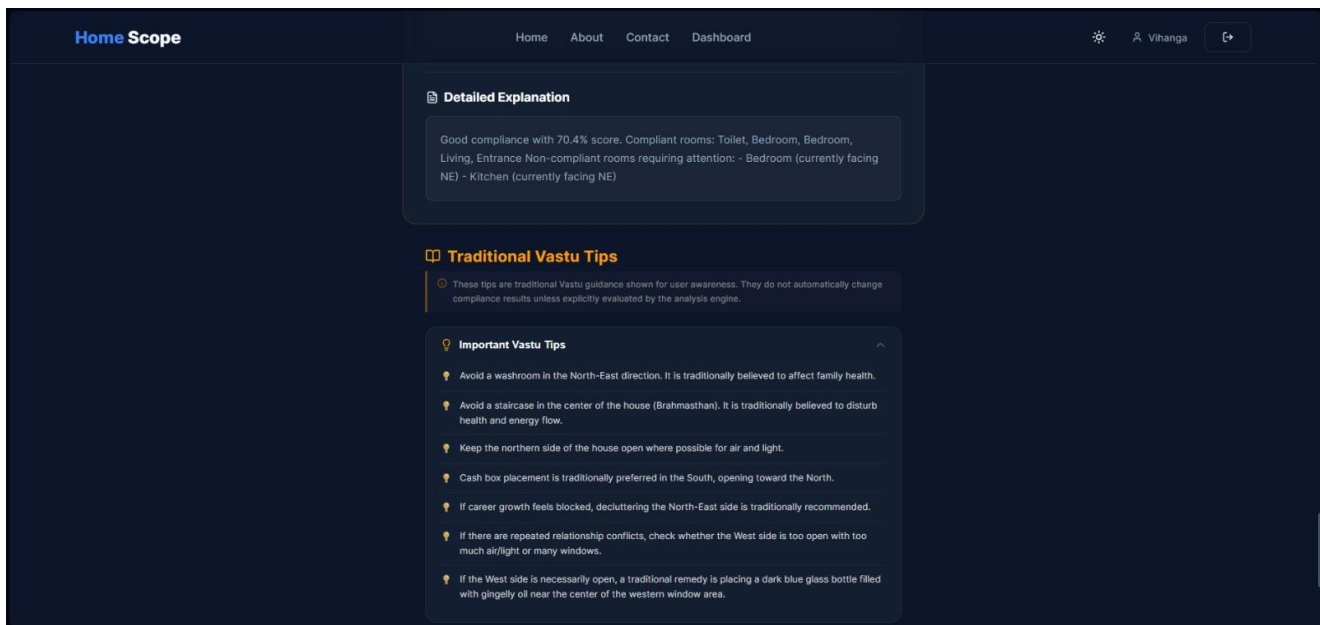


Figure 20: Detailed Explanation and Vastu Tips UI 1

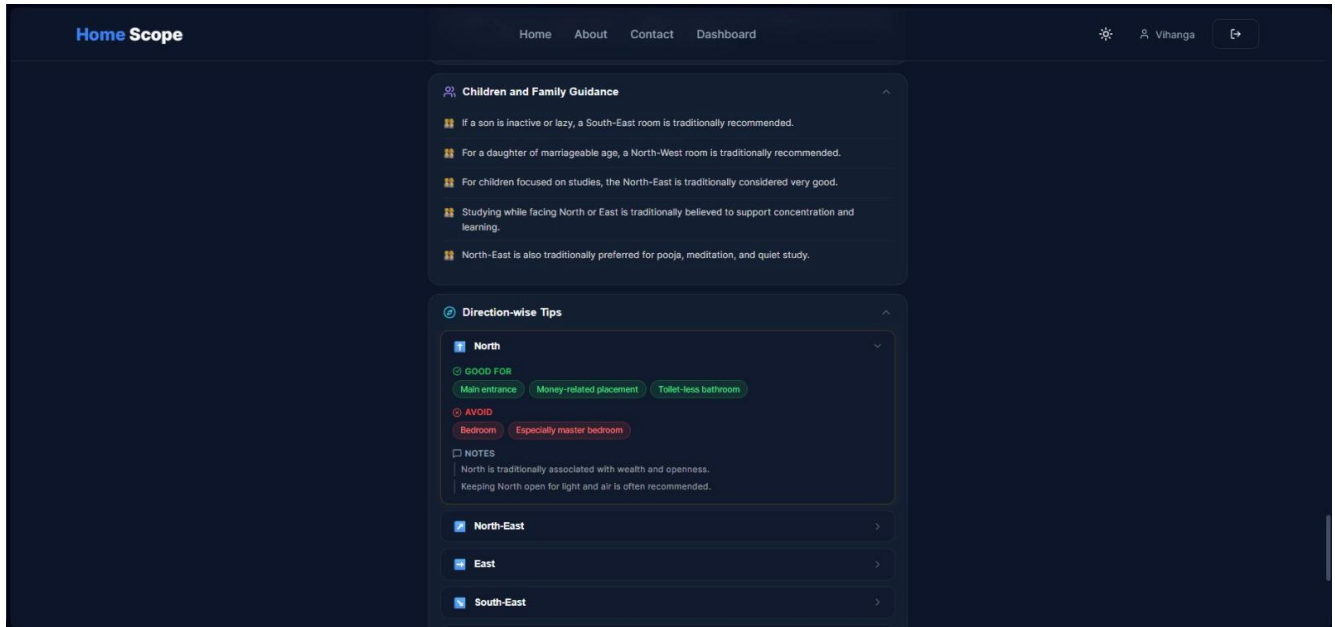


Figure 21: Detailed Explanation and Vastu Tips UI 2

Overall, the results demonstrate that the proposed system is capable of performing automated floor plan analysis and Vastu evaluation with a reasonable level of accuracy. While there are limitations in detection performance for certain classes, the system successfully achieves its primary objective of providing AI-based Vastu compliance analysis.

3.2 Research Findings and Discussions

The findings of this study highlight both the effectiveness and the limitations of applying deep learning techniques to floor plan analysis and Vastu compliance evaluation. The results demonstrate that the proposed system is capable of performing automated detection and rule-based reasoning; however, the performance is influenced by several technical and data-related factors.

One of the key findings is that the YOLOv8 model performs reasonably well in detecting architectural elements that have clear visual patterns and consistent representations across the dataset. Classes such as living rooms, doors, entrances, and windows showed relatively stronger detection performance. This can be attributed to the fact that these elements are visually distinct and frequently present in most floor plan images, allowing the model to learn their features effectively.

On the other hand, the system exhibited weaker performance for classes such as toilets, staircases, and master bedrooms. These elements often vary significantly in shape, size, and labeling conventions across different floor plans. In some cases, these elements may also be represented with minimal or ambiguous visual cues, making them harder for the model to detect accurately. This finding indicates that object detection performance is highly dependent on the consistency and diversity of the training dataset.

Another important observation is related to the size of the dataset used for training. With only 501 images, the dataset is relatively small for training a deep learning model. Although transfer learning using the pretrained yolov8m model helped improve performance, the limited dataset size still affects the generalization capability of the model. This is reflected in the moderate values of precision (0.438) and recall (0.455), as well as fluctuations observed in the training graphs.

The analysis of training and validation curves suggests that the model learns effectively during the initial epochs, but performance stabilizes and fluctuates afterward. This behavior indicates that the model reaches its learning capacity given the available data, and further improvement would require either a larger dataset or additional data augmentation techniques.

The results also highlight the importance of preprocessing techniques. The use of CLAHE, bilateral filtering, and edge detection contributed to improving the clarity of floor plan images, which in turn supports better detection performance. However, preprocessing alone cannot fully compensate for variations in drawing styles and symbol representations found in real-world floor plans.

From a system perspective, the integration of object detection with spatial reasoning proved to be effective. The use of contour detection to estimate the house boundary and calculate the Brahmasthan enabled the system to perform central region analysis. Similarly, the use of the atan2 function for direction calculation allowed accurate mapping of detected elements relative to the user-defined North direction.

However, a critical dependency of the system is the accuracy of user input, particularly the selection of the North direction. Incorrect orientation input directly affects all directional calculations and Vastu evaluations, which may lead to misleading results. This highlights a limitation of semi-automated systems that rely on user interaction.

The rule-based Vastu evaluation engine successfully translates traditional Vastu principles into computational logic. The system is capable of generating compliance scores, identifying violations, and providing recommendations. Nevertheless, the rule set implemented in the system represents only a subset of Vastu guidelines. Since Vastu Shastra includes a wide range of interpretations and regional variations, the system cannot be considered a complete representation of all possible rules.

Another key finding is that the system performs well as a decision-support tool rather than a decision-making system. The generated results are informative and helpful for users, but they should not be treated as absolute or authoritative conclusions. This aligns with the ethical considerations discussed earlier, where the system is intended to assist users rather than replace expert judgment.

From a usability perspective, the web-based implementation using React and FastAPI provides a smooth and responsive user experience. The integration of visualization tools such as Konva enhances the interpretability of detection results, allowing users to clearly understand how the system identifies and evaluates different elements within the floor plan.

Overall, the findings confirm that the proposed system successfully achieves its primary objective of automating Vastu-based floor plan analysis using AI techniques. The combination of YOLOv8-based object detection, image preprocessing, spatial reasoning, and rule-based evaluation forms an effective pipeline for analyzing residential layouts.

However, the discussion also emphasizes that further improvements can be made by increasing the dataset size, enhancing model training, refining detection accuracy for weak classes, and expanding the rule-based

engine to cover a broader range of Vastu principles. These improvements would significantly enhance the reliability and applicability of the system in real-world scenarios.

4. CONCLUSION

The aim of this project was to develop an AI-Based Vastu Detection and Compliance Evaluation system capable of analyzing 2D floor plan images and generating Vastu-based insights using automated techniques. The system successfully integrates deep learning, image processing, and rule-based reasoning to provide a functional and user-friendly solution for residential layout evaluation.

The implementation of the YOLOv8 object detection model enabled the system to identify key architectural elements such as rooms, doors, and windows from floor plan images. The use of preprocessing techniques, including CLAHE, bilateral filtering, and edge detection, contributed to improving image quality and detection performance. Additionally, spatial analysis methods such as contour detection and direction calculation using the atan2 function allowed the system to interpret positional relationships between detected elements.

The rule-based Vastu evaluation engine further extends the functionality of the system by converting traditional Vastu principles into computational logic. As a result, the system is capable of generating a compliance score, performing room-wise analysis, and providing practical recommendations to users. The inclusion of a manual input option also ensures flexibility when automatic detection is not sufficient.

The experimental results indicate that the model achieves moderate performance, with a precision of 0.438 and recall of 0.455. While the system demonstrates strong performance in detecting elements such as doors, windows, and living rooms, certain classes such as toilets and staircases show lower accuracy. These limitations are primarily influenced by the size and diversity of the dataset used for training.

Despite these challenges, the system successfully fulfills its primary objective of providing an automated and accessible approach to Vastu analysis. The web-based implementation ensures that users can easily upload floor plans, interact with the interface, and obtain meaningful insights without requiring expert-level knowledge.

The study also highlights several opportunities for future improvement. Increasing the dataset size, enhancing model training, and incorporating more advanced detection techniques could improve overall accuracy. Furthermore, expanding the rule-based engine to include a wider range of Vastu principles would enhance the system's applicability. Additional features such as automatic orientation detection and improved boundary estimation methods could further strengthen the reliability of the system.

In conclusion, the proposed AI-based system demonstrates the feasibility of combining deep learning and

traditional architectural principles to create an intelligent analysis tool. The system provides a strong foundation for future research and development in automated floor plan interpretation and Vastu-based decision support systems.

5. REFERENCES

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